Band-1 Hadamard Order 8516 — Miyamoto (Double Paley-I Seed H_2128 from H_1064 q=1063 + C_2130); Miyamoto-only NOT in SPK; Gap 2052 Still OPEN

H_1064 Paley-I PASS; H_2128 Double PASS; C_2130 PASS; H_8516 Miyamoto PASS; classical SPK FAIL → Miyamoto-only NOT in SPK; gap 2052 still OPEN; conjecture OPEN; GATE-04 LOCKED

C.M. Baird; ZoraASI mission-control Band-1 dash

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Document class. Scientific computational / construction certificate (Band-1).
Program. Open-mathematics collapse list · Human+AI attack order · ZoraASI mission-control.
Parents. `2026-09-16_Band1_Hadamard_8452_Miyamoto`; `2026-09-16_Band1_Hadamard_8356_PaleyII`; `2026-09-15_Band1_Hadamard_6628_Miyamoto`.
Lab. `/workspace/experiments/band1/hadamard_8516_miyamoto/`.
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Honesty stamp (non-negotiable).
This memo claims **finite constructions of orders 1064, 2128, and 8516 only** (verified $\max|HH^\top - nI| = 0$), via Miyamoto with Double(Paley-I) seed H_{2128} = Sylvester-double(H_{1064}), H_{1064} = Paley-I($q = 1063$), and symmetric Paley conference C_{2130} over GF(2129) ($q = 2129 \equiv 1 \pmod 4$). Classical SPK for 8516: Paley-I **FAIL**s ($q = 8515 = 5 \cdot 13 \cdot 131$ composite); Paley-II **FAIL**s ($q = 4257 = 3^2 \cdot 11 \cdot 43$ NOT prime power — no matrix attempt); Kronecker **FAIL**s (4×2129). Therefore order **8516 is a Miyamoto-only bonus NOT in SPK**. It **does not** claim the full Hadamard conjecture. It **does not** stamp Clay-closed or finished-ToE. GATE-04 collection remains **LOCKED**. No RH / P≠NP / YM / Frankl / NS-solved / AC-solved / Hadamard-conjecture-solved claims. Standard constructive gap **2052** remains **OPEN** under SPK. `clay_closed = false`. Prior ship: **8452 Miyamoto-only** (seed **2112** folded). Prior SPK seeds **2112 / 2088+8356 / 2028+8116 / 1996 / 1932 / 7604 / 1876+7508 / 1872 / 1860 / 7444 / 1848 / 7396 / 1788 / 888 / 1776 / 860 / 1720 / 6884 / 1708 / 840 / 1680 / 6724 / ...** retained; Miyamoto-only bonuses **8516/8452/7988/7732/7492/7156/7108/6836/6788/6676/6484/6404/6388/6196/...** excluded; **1064+2128** folded into ONBOX/SPK when verified; **8516** never folded (Miyamoto alone never folds SPK). Alternate Kronecker 4×532 for seed: **FAIL/not eligible** (532 NOT in ONBOX).

Abstract

Finite constructions of orders **1064**, **2128**, and **8516**.

- H_{1064} = Paley-I ($q = 1063$) — Double half
- H_{2128} = Sylvester-double(H_{1064}) — preferred Miyamoto seed (Kronecker 4×532 FAIL/not eligible; Paley-I direct $q = 2127$ FAIL; Paley-II $q = 1063$ FAIL)
- C_{2130} = Paley conference over GF(2129), $q = 2129$
- H_{8516} = Miyamoto(H_{2128}, C_{2130})
- Paley-I for 8516: **SKIP** ($q = 8515$ composite)
- Paley-II for 8516: **SKIP** ($q = 4257$ NOT prime power; no matrix attempt)
- Kronecker for 8516: **SKIP** ($4 \cdot 2129$ not admissible)

SPK: 8516 Miyamoto-only NOT in SPK. Seeds 1064+2128 enter SPK (Paley-I+Double). Prior SPK **2112 / 2088+8356 / 2028+8116 / 1996 / ...** retained.
Gap 2052: OPEN. **Conjecture:** OPEN. **clay_closed:** false. **GATE-04:** LOCKED.

`python3 verify_hadamard_8516.py`

Results. overall_pass=true; units 152/152; wall 507.8963s; all max|HHˆT-nI|=0.

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Arithmetic

- Target $n = 8516 = 4 \cdot 2129$; 2129 prime $\equiv 1 \pmod 4$ → Miyamoto-eligible.
- Need H_{2128} and C_{2130} (Paley conference over prime field $q = 2129$).
- Seed: Double(Paley-I) half=1064, $q = 1063$ (prime $\equiv 3 \pmod 4$) → H_{1064} → Sylvester-double → H_{2128} (Kronecker 4×532 FAIL — 532 not ONBOX; Paley-I direct $q = 2127$ FAIL; Paley-II $q = 1063$ FAIL).
- SPK: Paley-I $q = 8515$ **FAIL**; Paley-II $q = 4257$ **FAIL** (NOT prime power); Kronecker **FAIL** → **Miyamoto-only NOT in SPK**.
- Prior: 8452 Miyamoto-only shipped; this ship $q = 2129 \rightarrow n = 8516$.

Constructions (verified)

Object	Method	max|HHˆT-nI|	Wall (s)
H_1064 Paley-I	Paley-I q=1063 Double half	0	1.121
H_2128 Double	Sylvester-double(H_1064)	0	4.2898
C_2130	Paley conf GF(2129) q=2129	0	6.3968
H_8516 Miyamoto	Miyamoto(H_2128 Double, C_2130)	0	490.4205
H_8516 Paley-I	SKIP q=8515=$5 \cdot 13 \cdot 131$	—	—
H_8516 Paley-II	SKIP q=4257=$3^2 \cdot 11 \cdot 43$ (no matrix)	—	—
H_8516 Kronecker	SKIP $4 \cdot 2129$ not admissible	—	—

Object	Method	max|HH^T-nI|	Wall (s)

SPK closure note

Seeds include prior SPK ships through **8452** lineage (**2112**; **2088+8356**; **2028+8116**; **1996**; **1932**; **7604**; **1876/7508**; **1872**; **1860/7444**; **1848/7396**; **1788**; **888/1776**; **860/1720/6884**; **1708**; **6724**; **840/1680**; **...**; **NOT** 8516/8452/7988/7732/7492/7156/7108/6836/6788/6676/6484/6404/6388/6196 + verified **1064+2128** (Paley-I+Double); **exclude** Miyamoto-only bonuses including **8516**. Known count 1668. Next constructive gap **2052** (OPEN).

Quantity	Value
8516 in SPK	**False** (Miyamoto-only)
2128 in SPK	**True** (Double Paley-I)
1064 in SPK	**True** (Paley-I q=1063)
2112 / 2088 / 8356 in SPK	**True** (prior)
8516 / 8452 / 7988 / 7732 in SPK	**False** (Miyamoto-only; excluded)

Honesty gates

- `clay_closed = false`
- GATE-04 collection **LOCKED**; no invented J
- Finite-order constructions **1064**, **2128**, and **8516** ONLY (verified max|HH −nI|=0)
- **NO** full Hadamard conjecture claim; **NO** finished-ToE; **NO** RH / P NP / YM / Frankl / NS-solved / AC-solved
- SPK membership only via classical Paley-I / Paley-II / Kronecker (**Miyamoto alone is NOT SPK**)
- Standard constructive gap **2052** remains **OPEN**

Next optional

- **q=2141→n=8564 PaleyII(q=1069) OR q=2161→n=8644 Kronecker(4x540) OR q=2209→n=8836 PaleyI(q=2207)+Double(half=1104) OR method-attack SPK gap 2052 OR Track B / Object 1 meter await**